

# Homework 5

Name Here

## A model for bee toxicity

Again consider the bee toxicity problem from the previous homework. The data on toxicity levels and hive collapse is given again below for convince.

| x    | y |
|------|---|
| 1.06 | 0 |
| 1.41 | 1 |
| 1.85 | 1 |
| 1.50 | 1 |
| 0.46 | 0 |
| 1.21 | 1 |
| 1.25 | 1 |
| 1.09 | 1 |
| 1.76 | 1 |
| 1.75 | 1 |

Assume that beehive collapse,  $y_i$ , given pollutant exposure level  $x_i$ , is  $Y_i \sim \text{Bernoulli}(\theta(x_i))$ , where  $\theta(x_i)$  is the probability of death given dosage  $x_i$ . We will assume the *probit* regression model,  $\theta_i(x_i) = \Phi(\alpha + \beta x_i)$  where  $\Phi$  denotes the CDF of a standard normal distribution. Probit regression has a simple latent variable interpretation that is particularly amenable to Gibbs sampling. We will implement the Gibbs sampler by deriving the complete conditionals in a model where we have added additional augmented paramters,  $z_i$ .

**1a** Let  $z_i \sim N(\alpha + \beta x_i, 1)$ . Show that  $y_i = I[z_i > 0]$  (1 if  $z_i$  is greater than 0, and 0 otherwise) implies the probit model above.

**1b.** Derive the complete conditionals  $p(\alpha \mid z, y, x, \beta)$  and  $p(\beta \mid z, y, x, \alpha)$  assuming a flat prior on  $\alpha$  and  $\beta$ .

**1c** Now derive the complete conditionals of  $\alpha$  and  $\beta$  assuming instead assuming a priori  $\alpha \sim N(-10, \sigma = 5)$  and  $\beta \sim N(0, \sigma = 10)$  (note: standard deviations are provided not variances).

**1c** Derive the complete conditional of  $z_i$ ,  $p(z_i \mid z_{-i}, y, x, \alpha, \beta)$ . Find an R package or function that you can use to sample from this density and report the package you plan to use.

**1d.** Use the above conditionals, assuming the normal priors on  $\alpha$  and  $\beta$ , and implement and run a Gibbs sampler with multiple chains. Run the sampler long enough so that the effective sample size for all parameters is at least 1000 and discuss mixing. Report at 95% confidence interval for the LD50 and report the posterior probability that LD50 is greater than 1.2.